One Image, Twelve Mathematics: A Cross-Family Benchmark and the U-Shaped Editability of Image Representations

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***Abstract*—An image is always a matrix, but it can be re-expressed in many ways: as a sum over a Fourier, cosine, or wavelet basis, as a sparse combination of atoms from an overcomplete dictionary, as a handful of geometric primitives, as a small neural function of position, as an explicit closed-form formula, or as a code in the latent space of a generative model. We carry one image across twelve such families, on a common set of benchmark images and at a common parameter budget, and ask a question the fidelity literature does not: when you edit the parameters, what happens to the image? Two measured results and one finding follow. *First*, at an equal parameter budget the twelve families reconstruct comparably (peak signal-to-noise ratio 27–35 dB); fidelity does not distinguish them. *Second*, an editability-locality metric does: a single coefficient of a global transform is spatially spread (Fourier 0.16, DCT 0.23 concentration) while a coefficient of a space-localized transform is essentially local (wavelet 0.999, KLT 1.0). *The finding* is that editability is *U-shaped* across the abstraction spectrum: it is high at two poles, the designed-structure pole (geometric primitives, sparse atoms, space-localized transforms, where humans built local meaningful coordinates) and the learned-manifold pole (disentangled generative directions), and it collapses toward noise in between, where perturbing a raw neural-field weight or a constant of a brittle fitted formula destroys the image. A parameter is both stable and meaningful only when it indexes a low-dimensional manifold of plausible images with locally disentangled coordinates. We give the twelve families with their governing equations, the cross-family benchmark on six images, and the honest scope: arbitrary photographs do not reduce to compact, faithful, human-readable equations, and the results say where a representation's parameters are, and are not, a usable handle.**

***Index Terms*—image representation, sparse coding, wavelets, KLT, implicit neural representation, Gaussian splatting, symbolic regression, editability, rate-distortion, interpretability, reproducibility.**

I. INTRODUCTION

An image is a matrix, but it is rarely stored or reasoned about as one. Compression writes it in an orthonormal basis; a generative model writes it as a latent code; vector graphics write it as shapes; parametric-equation art writes it as a formula. Each of these is a *representation*, a way to write an image as parameters, and each promises something different about what those parameters mean. The forward question, how faithfully a representation reconstructs the image at a given budget, is well studied. The question we ask is the inverse one: when you move the parameters, does the image move in a stable, meaningful way, or does it fall apart?

We carry one image across twelve families of mathematical representation (Section II) and measure two things on a common set of benchmark images at a common parameter budget (Section III): fixed-budget fidelity, and an editability-locality metric. The result is that fidelity is comparable across families and does not distinguish them, whereas editability does, and it is organized as a U-shape across the abstraction spectrum (Section IV). The contributions are: (i) a single-image, twelve-family benchmark with each family's governing equation; (ii) the measured editability-locality contrast between global and space-localized representations; (iii) the U-shaped editability finding and its honest scope.

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Preprint, 2026. Code, the committed derived artifacts, and the interactive lab are released under MIT at https://github.com/fsantibanezleal/CAOS_IMGLAB. Every external engine and reference carries its own license and citation; the explicit formulas shown are the author's own.

II. TWELVE REPRESENTATION FAMILIES

Orthonormal transforms. Fourier, DCT [1] and wavelets [2] write the image in a fixed orthonormal basis, $x = \sum_k c_k \phi_k$ with $c_k = \langle x, \phi_k \rangle$; the map is exact and invertible, and keeping the top-k coefficients gives the classical rate-distortion tradeoff [3]. The KLT (PCA) replaces the fixed basis with the data-optimal basis learned from image patches. A Fourier coefficient is a global grating; a DCT coefficient a per-block frequency; a wavelet coefficient is local in space and scale.

Overcomplete frames and sparse dictionaries. Dropping orthonormality, a dictionary D with more atoms than pixels represents the image with a sparse code, $\min \|a\|_0$ s.t. $x \approx Da$, solved greedily by matching pursuit [4] on a learned or overcomplete dictionary [5]. A related family uses **Gabor atoms**, oriented Gaussian-windowed cosines [6], each a legible object (position, width, orientation, frequency, phase). Editing is local and semantic per atom, the first pole of high editability.

Geometric primitives. The image is approximated by an ordered list of translucent ellipses fit greedily to the residual; each shape is an independent, meaningful, local coordinate.

Implicit neural field. A small coordinate network (SIREN [7], with Fourier features [8]) fits $f_\theta(x, y) \approx \text{image}(x, y)$; the image becomes the weights θ, continuous and resolution-free, but the parameters are entangled, no single weight is a named handle.

Symbolic closed-form. Each channel is fit as an explicit trigonometric formula by ridge regression on D random

Fourier features, $\mathrm{ch}(x, y) = a_0 + \sum_k [a_k \cos(\omega_k \cdot (x, y)) + b_k \sin(\omega_k \cdot (x, y))]$, linear in known analytic basis functions so the picture literally is a readable equation [9]. Fidelity is honest and bounded: a smooth gradient reaches $\approx 57\,\mathrm{dB}$ from 512 terms, a hard-edged checkerboard only ≈ 14, because a finite trigonometric sum cannot render a discontinuity.

Gaussian mixture (2D splatting). The image is a sum of coloured anisotropic Gaussians, $\mathrm{ch}(x, y) = b + \sum_k c_k \exp(-\frac{1}{2} d_k^\top \Sigma_k^{-1} d_k)$, optimized jointly, the image counterpart of the representation behind real-time radiance fields [10], [11].

Thin-plate radial basis. A linear combination of a fixed thin-plate kernel plus an affine plane, $\mathrm{ch}(x, y) = a_0 + a_1 x + a_2 y + \sum_i w_i\, \phi(\|(x, y) - c_i\|)$ with $\phi(r) = r^2 \log r$, the smoothest interpolant of scattered data [12], solved in closed form.

Chebyshev series. A truncated tensor polynomial series $\sum_{ij} a_{ij} T_j(x) T_i(y)$, $T_k(t) = \cos(k \arccos t)$, fit live in the browser via discretely orthonormalized moments [13]; fidelity saturates on hard edges, the classical moments tradeoff.

Fourier descriptors. The dominant closed contour is traced and transformed into a sum of rotating circles $z(t) = \sum_k c_k e^{i f_k t}$, an exact orderable parametric description of a one-dimensional curve.

Generative latent. The image is a code in the latent space of a variational autoencoder [14]; the useful directions are disentangled and semantic, the second pole of high editability.

III. Cross-family benchmark

We bake each representation of six benchmark images (a photograph, a painting, a mathematical-art image, an astronomical image, a texture, and a synthetic gradient) at a common 256×256 resolution and evaluate two axes (Fig. 1). *Fixed-budget fidelity*: at an equal parameter budget (the top 5% of transform coefficients, ≈ 3300 numbers, matched against ≈ 90 ellipses, ≈ 2300 neural weights, 250 Gabor atoms, 300 Gaussians, 225 thin-plate kernels, 512 trigonometric terms, or a $4 \times 32 \times 32$ latent), the twelve families land within 27–35 dB (wavelet 34.8, DCT 33.2, neural field 33.0, Fourier 32.3, Gabor 32.1, primitives 32.0, Gaussian mixture 31.9, thin-plate 29.4, symbolic 29.4, KLT 29.4, Chebyshev 28.9, VAE latent 26.8). Fidelity does not separate the families. *Rate-distortion*: over coefficient fractions from 0.5% to 25% the space-localized wavelet dominates (up to $49.4\,\mathrm{dB}$ at 25%), then DCT and Fourier, with the patch KLT lower, the classical energy-compaction ordering. *Editability-locality*: we measure how spatially concentrated the effect of a single coefficient is; the global transforms are non-local (Fourier 0.156, DCT 0.228) while the space-localized transforms are local-exact (wavelet 0.999, KLT 1.0). This is the axis on which the families genuinely differ.

IV. The finding: U-shaped editability

Putting the two axes together yields the finding (Fig. 2). Because fixed-budget fidelity is comparable, what distinguishes the families for a user who wants to *edit* the image is whether a parameter is a stable, meaningful handle, and that is U-shaped across abstraction. Editability is high at the designed-structure pole, geometric primitives (move one ellipse and only that region changes), sparse and Gabor atoms (each a legible local object), and space-localized transforms (wavelet and KLT, locality ≈ 1), because humans built local, meaningful coordinates. It is high again at the learned-manifold pole, the disentangled directions of a generative latent space, because the model learned a low-dimensional manifold of plausible images with semantic axes. And it collapses in between: a single coefficient of a global transform is spatially spread (locality 0.16–0.23, so a local edit is impossible), a raw neural-field weight is entangled (no weight is a named handle), and a constant of a brittle fitted formula is fragile (perturbing it destroys the image, exactly because a finite trigonometric sum has no local support). The organizing statement is that a parameter is both stable and meaningful only when it indexes a low-dimensional manifold of plausible images with locally disentangled coordinates; the designed and the learned poles achieve this by construction and by training, and the entangled middle does not.

V. Discussion and honest scope

The quantitative editability-locality is measured for the four transform families, where the local-versus-global contrast is exact and unambiguous; the U-shape across all twelve families is the organizing thesis, supported by that measurement, by the qualitative locality of each family's parameters, and by the fixed-budget benchmark that rules out fidelity as the differentiator. The result is descriptive, not a new representation: it says where a representation's parameters are, and are not, a usable handle, which is the practical question behind interpretable and editable image models. The honest scope is explicit: arbitrary photographs do not reduce to compact, faithful, human-readable equations, the symbolic and formula families make genuine but bounded partial reconstructions (a smooth gradient fits to $\approx 57\,\mathrm{dB}$, a hard edge to ≈ 14), and the benchmark reports those partial results rather than overclaiming. The generative-latent pole is argued from the disentanglement literature and the family's construction rather than from a locality number, and is labelled as such.

VI. Conclusion

We carried one image across twelve mathematical representation families on a common benchmark and found that equal-budget fidelity does not distinguish them (27–35 dB) while editability does, and that editability is U-shaped across the abstraction spectrum, high at the designed-structure and learned-manifold poles and low in the entangled middle, with the transform families' editability-locality measured directly (global 0.16–0.23 versus local-exact ≈ 1). A parameter is a usable handle only when it indexes a low-dimensional manifold with locally disentangled coordinates. All code and committed artifacts are released so the benchmark and the figures are reproducible.

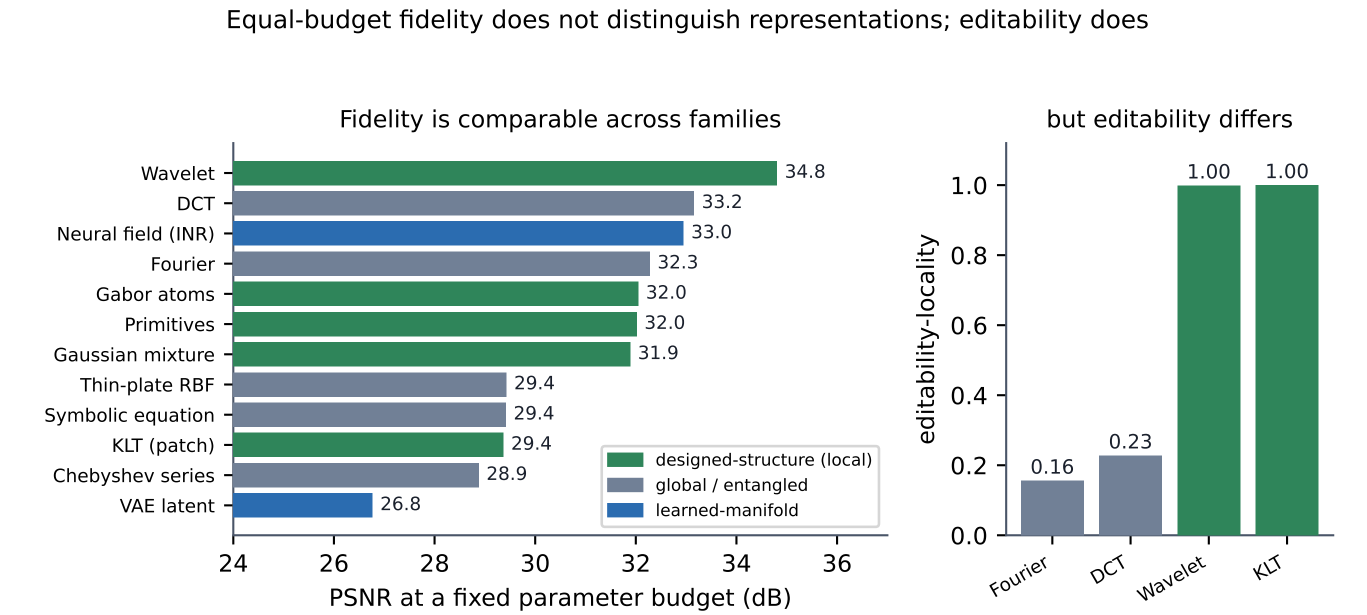

Fig. 1. The benchmark, read from the committed `_bench/index.json`. Left: peak signal-to-noise ratio at a fixed parameter budget across the twelve families; fidelity is comparable (27–35 dB) and does not distinguish them, coloured by pole (designed-structure local, global or entangled, learned manifold). Right: the editability-locality metric for the four transform families; a single coefficient of a global transform is spatially spread (Fourier 0.16, DCT 0.23) while a coefficient of a space-localized transform is essentially local (wavelet 0.999, KLT 1.0).

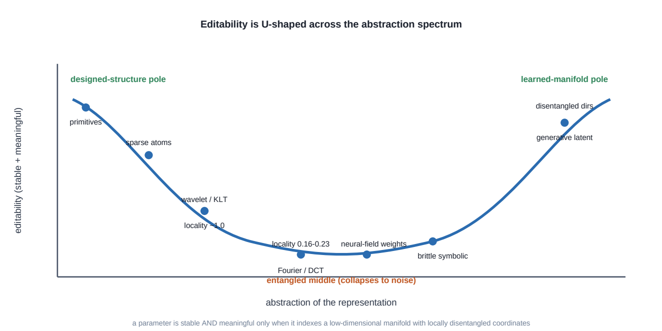

Fig. 2. Editability is U-shaped across the abstraction spectrum: high at the designed-structure pole (primitives, sparse atoms, space-localized transforms) and the learned-manifold pole (disentangled generative directions), and low in the entangled middle (global transforms, raw neural-field weights, brittle fitted formulas).

Data and Code Availability

Code, the committed benchmark artifacts, and the interactive lab are at https://github.com/fsantibanezleal/CAOS_IMGLAB under the MIT license. Benchmark images are public-domain or permissively licensed, plus procedurally generated license-clean images; each carries its citation in the repository. The figures are regenerated deterministically from the committed `_bench/index.json` by `manuscripts/image-representation-editability/figures/make_figs.py`.

Appendix A
Fixed-Budget Fidelity Table

At the matched budget the peak signal-to-noise ratios are: wavelet 34.8, DCT 33.2, neural field (SIREN) 33.0, Fourier 32.3, Gabor atoms 32.1, primitives 32.0, Gaussian mixture 31.9, thin-plate RBF 29.4, symbolic equation 29.4, KLT (patch) 29.4, Chebyshev series 28.9, VAE latent 26.8 dB, over six benchmark images (photograph, painting, mathematical art, astronomy, texture, synthetic gradient). The full per-image rate-distortion curves and the editability-locality values are committed in `data/derived/_bench/index.json`.

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